

## Progress Report 5

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**Project:** Smart Door Anomaly Detection & Cybersecurity for Elderly Care

**Course:** CSIS 4495 – Applied Research Project, Section 3

**Reporting Period:** March 17 – March 23, 2026

**Report Date:** March 23, 2026

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### Work Date / Hours Logs

| Date         | Hours | Description of Work Done                                                                                                                                                                                                                                                                                               |
|--------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mar 17, 2026 | 2.0   | Added runtime configuration for fall detector mode selection (rules / LSTM) in Implementation/config.py. Updated app startup in Implementation/api/__init__.py to support mode selection and fallback behavior. Added Implementation/models/model_info.json to store model metadata details.                           |
| Mar 18, 2026 | 2.5   | Updated Implementation/api/fall_detection_routes.py to improve fall route behavior and status response payload. Updated Implementation/frontend/lib/api.ts so frontend typings match backend status fields.                                                                                                            |
| Mar 19, 2026 | 2.0   | Added new fall route test coverage in Implementation/tests/test_fall_detection.py to validate endpoint behavior and reduce regression risk.                                                                                                                                                                            |
| Mar 20, 2026 | 2.0   | Fixed backend “Falls Today” counting in Implementation/api/routes.py using timezone-aware datetime parsing to avoid date mismatch issues.                                                                                                                                                                              |
| Mar 21, 2026 | 2.5   | Added Implementation/scripts/evaluate_lstm.py for LSTM evaluation workflow and Implementation/scripts/system_health_check.py for pre-run checks. Added Implementation/models/lstm_eval_report.txt artifact file. Updated Implementation/scripts/fall_detection_camera.py with runtime options and integration support. |
| Mar 23, 2026 | 1.0   | Performed testing and debugging of fall detection flow in both rules and LSTM modes. Verified event logging behavior and reviewed handling for visibility-related false alerts. Re-checked dashboard/API                                                                                                               |

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|--|--|-----------------------------------------------------------------------------------|
|  |  | consistency after timestamp-related fixes.   Team meet   Progress report creation |
|--|--|-----------------------------------------------------------------------------------|

**Total Hours: 12.0**

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### Summary Description of Work

This week, I focused on strengthening the fall detection module to improve usability, monitoring, and validation. I introduced configuration support to switch between rules-based and LSTM detection at runtime and enhanced the initialization flow to ensure safer model loading.

Additionally, I improved the fall detection API behavior and expanded the status response fields used by the frontend. To maintain consistency, I updated the frontend API typings accordingly. I also added test coverage for fall route behavior and completed debugging related to event logging and visibility-based edge cases.

For evaluation and operational readiness, I implemented scripts for LSTM performance evaluation and system health checks, along with generating an evaluation report artifact. I also resolved issues with date/time handling in backend fall statistics, ensuring the “Falls Today” metric is computed accurately.

Overall, this week’s work significantly improved the reliability, observability, and maintainability of the fall detection pipeline.

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### Repo Check-in of Implementation

The following files were updated/added during this reporting period:

- Implementation/config.py
- Implementation/api/\_\_init\_\_.py
- Implementation/models/model\_info.json
- Implementation/api/fall\_detection\_routes.py
- Implementation/frontend/lib/api.ts
- Implementation/tests/test\_fall\_detection.py
- Implementation/api/routes.py
- Implementation/scripts/evaluate\_lstm.py
- Implementation/scripts/system\_health\_check.py

- Implementation/models/lstm\_eval\_report.txt
- Implementation/scripts/fall\_detection\_camera.py

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## AI Use Section

| AI Tool        | Version / Account | Feature Used                                                                           | Value Added                                                                                                      |
|----------------|-------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| GitHub Copilot | Free Tier         | Code completion/suggestions while editing routes, scripts, and tests                   | Helped speed up boilerplate writing; all generated snippets were reviewed and adjusted before use.               |
| ChatGPT        | Pro Tier          | Design/debug guidance for fall API behavior, timestamp handling, and workflow planning | Used to validate implementation approach, identify edge cases, and organize weekly work into clear deliverables. |

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## Appendix: Prompt History

### Prompt 1:

“How should I design a runtime configuration system to switch between rules-based and LSTM fall detection without changing existing API behavior?”

### Prompt 2:

“What fields should be included in a fall detection status API response to support frontend monitoring and debugging?”

### Prompt 3:

“How can I evaluate an LSTM model for fall detection and generate a simple report for performance tracking?”